Exploratory Data Analysis (EDA) on CardioGoodFitness Dataset

Introduction

Exploratory Data Analysis (EDA) is the process of analyzing datasets to understand their structure, patterns, relationships, and key characteristics. The objective of this project is to perform EDA on the CardioGoodFitness dataset and derive meaningful insights.

Objectives

- Analyze the dataset and understand its features.
- Calculate descriptive statistics such as mean, median, and count.
- Identify missing values and outliers.
- Visualize data distributions.
- Study relationships between variables using correlation analysis.

Dataset Description

The CardioGoodFitness dataset contains customer information related to fitness products and usage patterns. The dataset includes variables such as Age, Education, Usage, Fitness, Income, and Miles.

Data Exploration

The dataset was loaded into Python using Pandas. Initial exploration was performed using functions such as:

- df.head()

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	TM195	18	Male	14	Single	3	4	29562	112
1	TM195	19	Male	15	Single	2	3	31836	75
2	TM195	19	Female	14	Partnered	4	3	30699	66
3	TM195	19	Male	12	Single	3	3	32973	85
4	TM195	20	Male	13	Partnered	4	2	35247	47

- df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   Product        180 non-null    object
 1   Age            180 non-null    int64 
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64 
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64 
 6   Fitness        180 non-null    int64 
 7   Income         180 non-null    int64 
 8   Miles          180 non-null    int64 
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
None
```

- df.describe()

```
              Age   Education       Usage     Fitness         Income  \
count  180.000000  180.000000  180.000000  180.000000     180.000000
mean    28.788889   15.572222    3.455556    3.311111   53719.577778
std      6.943498    1.617055    1.084797    0.958869   16506.684226
min     18.000000   12.000000    2.000000    1.000000   29562.000000
25%     24.000000   14.000000    3.000000    3.000000   44058.750000
50%     26.000000   16.000000    3.000000    3.000000   50596.500000
75%     33.000000   16.000000    4.000000    4.000000   58668.000000
max     50.000000   21.000000    7.000000    5.000000  104581.000000

            Miles
count  180.000000
mean   103.194444
std     51.863605
min     21.000000
25%     66.000000
50%     94.000000
75%    114.750000
max    360.000000
```

These functions helped in understanding the dataset structure and data types.

Missing Value Analysis

The dataset was checked for missing values using:

df.isnull().sum()

```
Product          0
Age              0
Gender           0
Education        0
MaritalStatus    0
Usage            0
Fitness          0
Income           0
Miles            0
dtype: int64
```

No significant missing values were found in the dataset.

Descriptive Statistics

Descriptive statistics were calculated using:

df.describe()

```
...            Age  Education       Usage     Fitness         Income  \
count  180.000000  180.000000  180.000000  180.000000     180.000000
mean    28.788889   15.572222    3.455556    3.311111   53719.577778
std      6.943498    1.617055    1.084797    0.958869   16506.684226
min     18.000000   12.000000    2.000000    1.000000   29562.000000
25%     24.000000   14.000000    3.000000    3.000000   44058.750000
50%     26.000000   16.000000    3.000000    3.000000   50596.500000
75%     33.000000   16.000000    4.000000    4.000000   58668.000000
max     50.000000   21.000000    7.000000    5.000000  104581.000000

            Miles
count  180.000000
mean   103.194444
std     51.863605
min     21.000000
25%     66.000000
50%     94.000000
75%    114.750000
max    360.000000
```

The analysis provided:

- Mean

```
print(df.mean(numeric_only=True))
```

```
Age             28.788889
Education       15.572222
Usage            3.455556
Fitness          3.311111
Income       53719.577778
Miles          103.194444
dtype: float64
```

- Median

```
...  Age            26.0
     Education      16.0
     Usage           3.0
     Fitness         3.0
     Income      50596.5
     Miles          94.0
     dtype: float64
```

- Minimum Value

```
...  Age              18
     Education        12
     Usage             2
     Fitness           1
     Income        29562
     Miles            21
     dtype: int64
```

- Maximum Value

```
Age              50
Education        21
Usage             7
Fitness           5
Income       104581
Miles           360
dtype: int64
```

- Standard Deviation

```
Age              6.943498
Education        1.617055
Usage            1.084797
Fitness          0.958869
Income       16506.684226
Miles           51.863605
dtype: float64
```

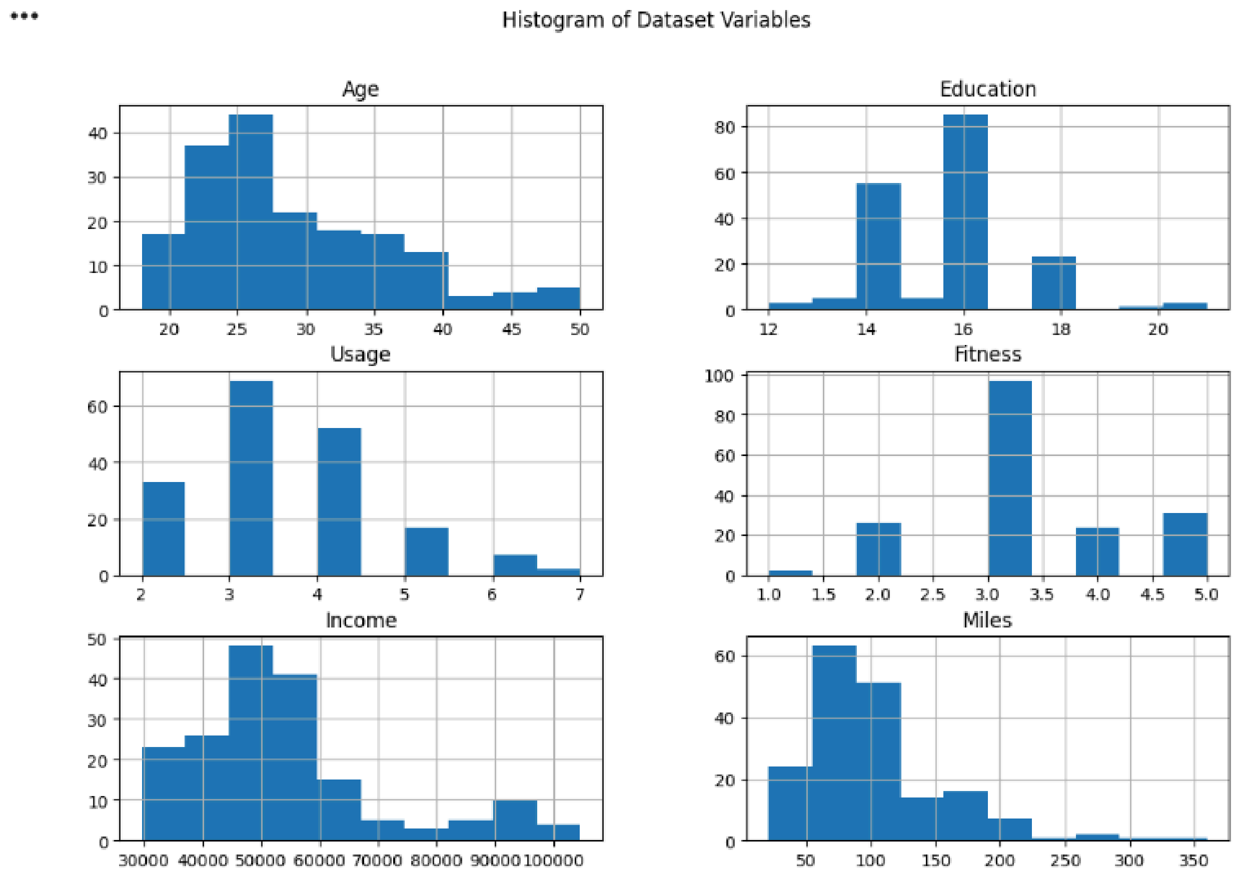
- Quartiles

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

These statistics helped summarize the numerical features of the dataset.

Data Visualization

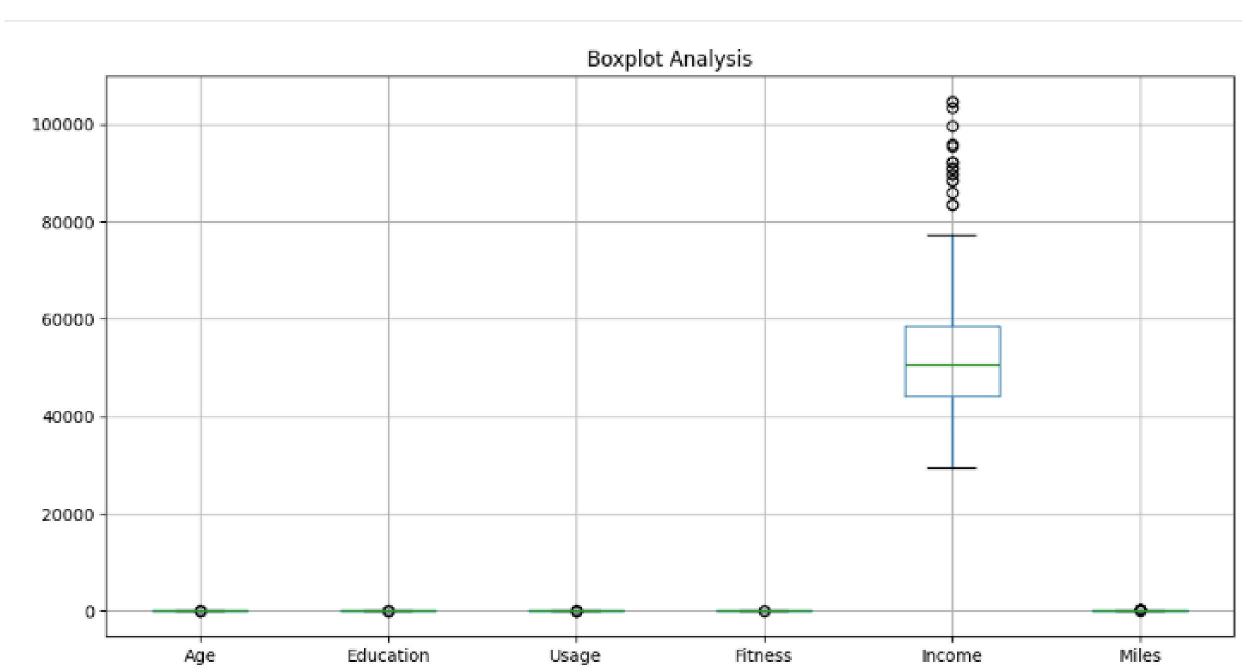
Histogram Analysis



Histograms were generated to understand the distribution of variables. The distributions showed variation among customer attributes such as age, income, and fitness levels.

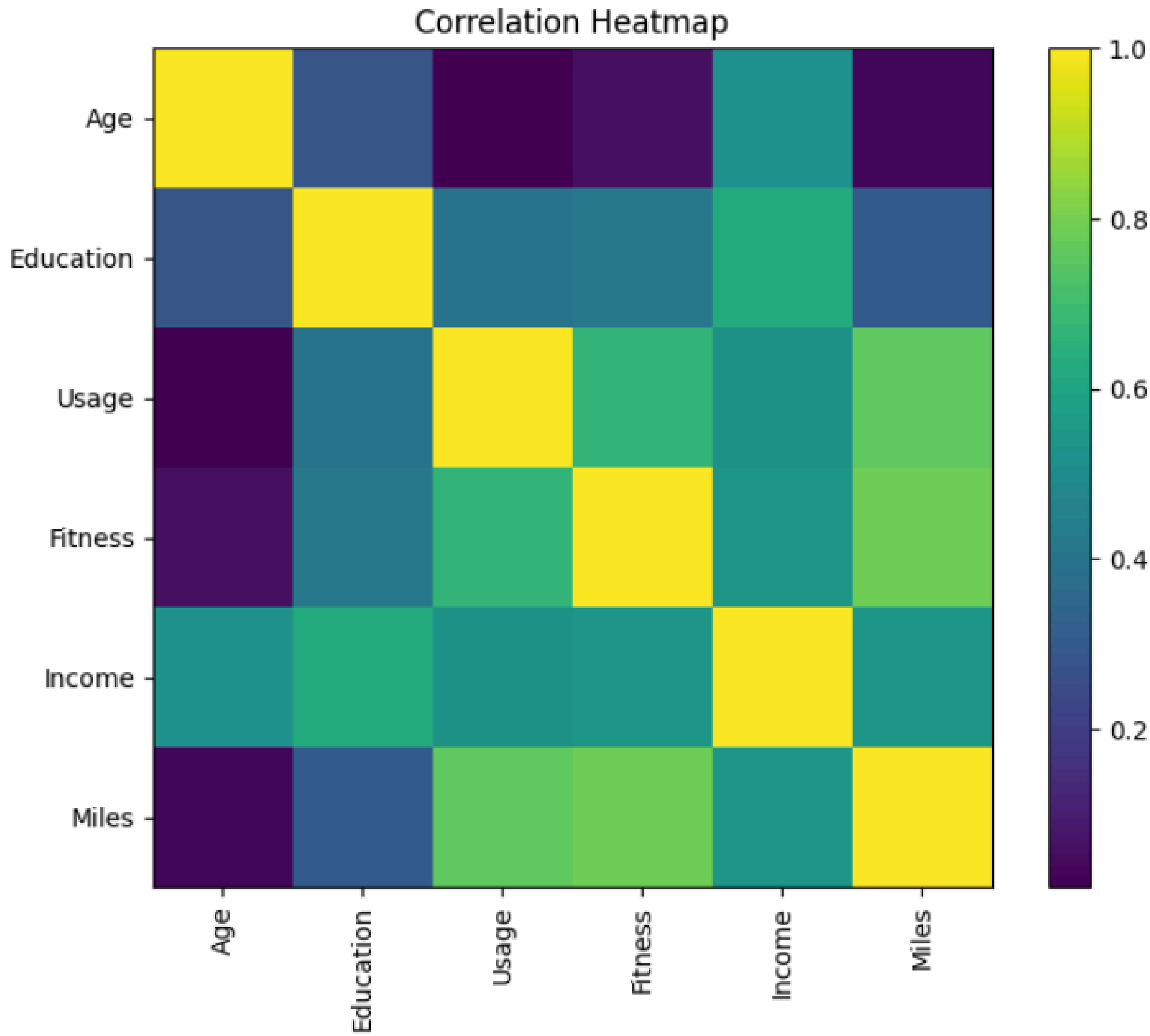
Boxplot Analysis

Boxplots were used to identify potential outliers. Some variables showed observations outside the normal range, indicating possible outliers.



Correlation Analysis

A correlation matrix and heatmap were generated to study relationships between variables.



Key Findings

- Usage and Miles showed a strong positive correlation.
- Fitness and Miles were positively correlated.
- Usage and Fitness showed a positive relationship.
- Income had a moderate positive relationship with fitness-related variables.
- Age showed relatively weak correlations with most variables.

Insights

The analysis revealed that customers who use fitness products more frequently tend to cover more miles and maintain higher fitness levels. Correlation analysis helped identify important relationships among customer attributes.

Conclusion

The Exploratory Data Analysis successfully identified patterns, trends, and relationships within the CardioGoodFitness dataset. The project demonstrated the importance of data exploration before advanced analysis or machine learning. The insights obtained can support data-driven decision-making and provide a better understanding of customer fitness behavior.